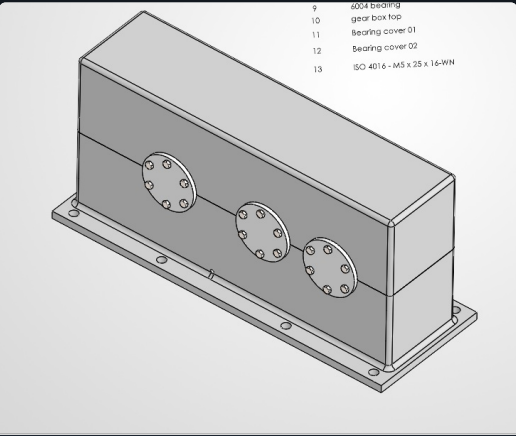
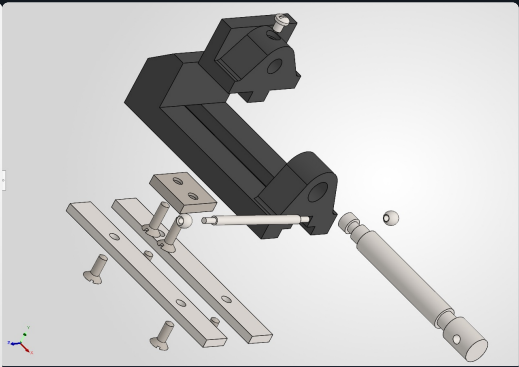


Mechanical Design Portfolio

FEATURED PROJECTS — GEARBOX & BENCH VISE ASSEMBLY



01 • Two-Stage Spur Gearbox
Gear design • Shafts • Bearings



02 • Bench Vise Assembly
Sliding jaw • Screw actuation

Modeled in
SolidWorks

Assembly • GD&T • BOM • Exploded
Views

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Project Overview

A two-stage spur gear reduction gearbox designed and modeled fully in SolidWorks, from individual gear and shaft components through to a complete bolted assembly. The project covers gear tooth design, shaft and bearing selection, housing design, and full assembly documentation including a bill of materials and exploded view.

2

REDUCTION STAGES

~6.1 : 1

OVERALL GEAR RATIO

14

UNIQUE COMPONENTS

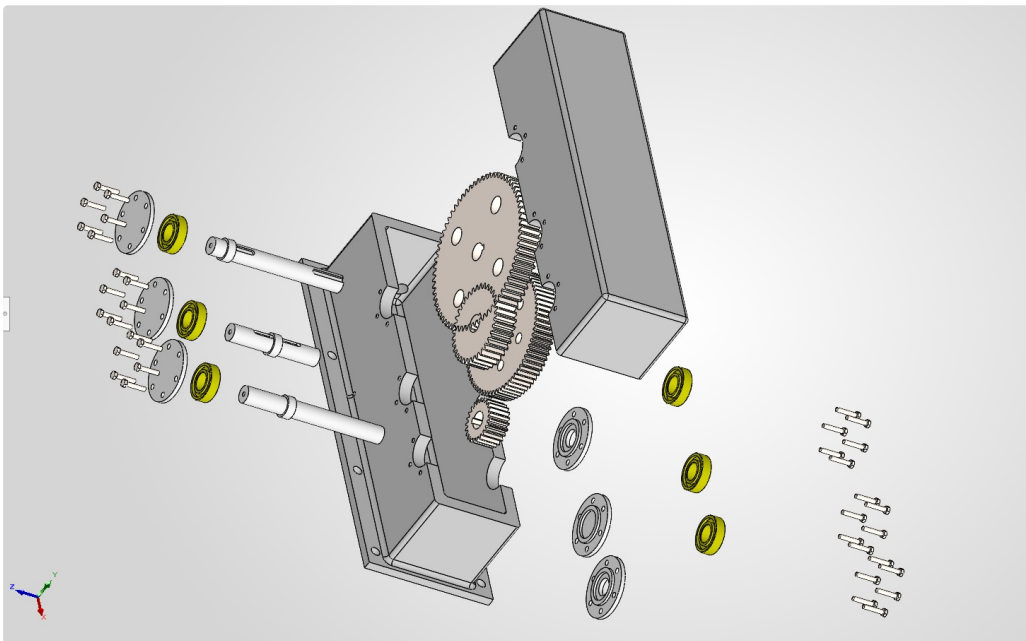
2.5

MODULE (MM), STAGE 2

Design Highlights

- **Two-stage reduction:** Stage 1 steps down through a 23T pinion driving a 69T gear (3.0:1); Stage 2 continues through a 34T to 69T pair, module 2.5 ($\approx 2.03:1$) — combining for an overall ratio of roughly 6.1:1.
- **Full assembly modeling:** housing (top and bottom halves), three shafts, gear set, and 6004 bearings modeled and mated as a functioning assembly with correct fits and clearances.
- **Serviceability:** bolted bearing covers (01 & 02) allow the shafts and bearings to be accessed without splitting the full housing.
- **Documentation:** complete bill of materials, exploded view, and fastener callouts (ISO 4016 – M5 hardware) generated directly from the assembly.

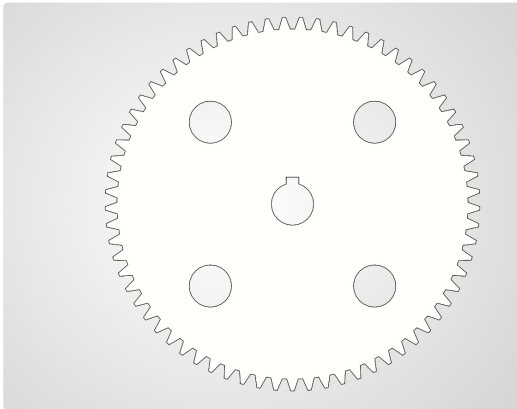
Exploded View



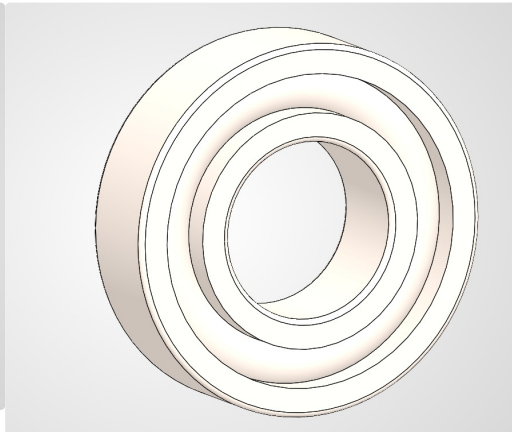
Full exploded assembly — shafts, gears, bearings, bearing covers, housing, and fasteners

Component Modeling

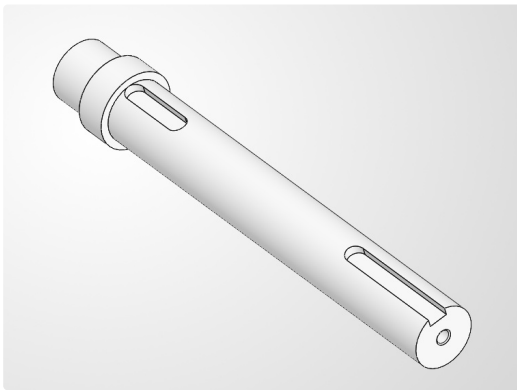
Every part in the assembly was individually modeled as a separate SolidWorks part, including correct gear tooth profiles, bearing geometry, and bolt-hole patterns, before being mated together.



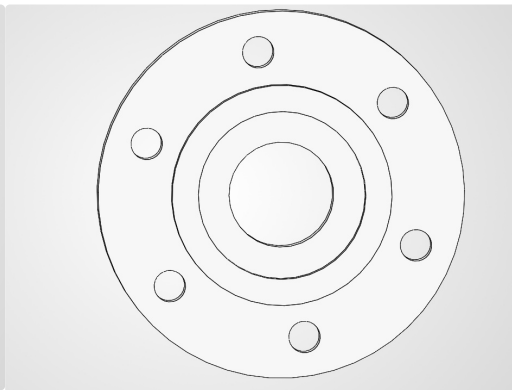
Spur gear — modeled tooth profile with bolt-hole pattern



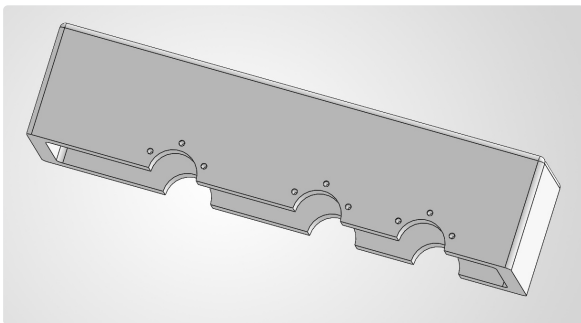
6004 deep-groove bearing



Gearbox shaft with retaining-ring grooves



Bearing cover — bolt pattern and bore fit



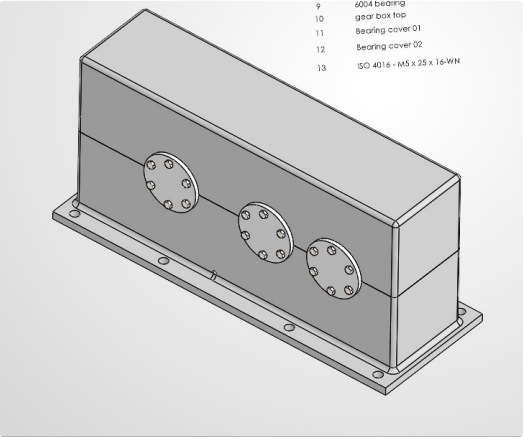
Housing (bottom half) — machined pockets for shaft and bearing seats

Bill of Materials

Extracted directly from the SolidWorks assembly BOM. Every component below was individually modeled as a separate part and mated into the final assembly.

ITEM	PART NUMBER / DESCRIPTION	QTY
1	Spur gearbox (housing, bottom half)	1
2	23T gear	1
3	69T gear	1
4	34T 2.5M gear	1
5	69T 2.5M gear	1
6	Gearbox (housing, secondary)	1
7	Gearbox shaft 2	1
8	Gearbox shaft 3	1
9	6004 bearing	6
10	Gearbox top (housing, top half)	1
11	Bearing cover 01	4
12	Bearing cover 02	2
13	ISO 4016 - M5 x 25 x 16-WN (fastener)	36

Assembled View



Final assembled gearbox housing

Project Overview

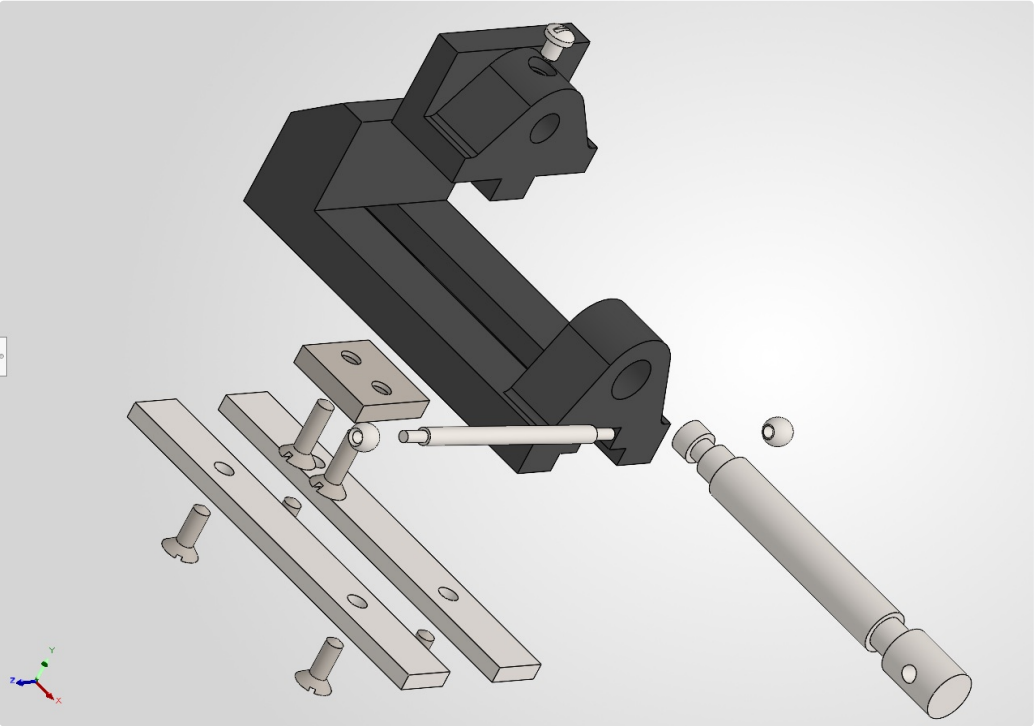
A machinist's bench vise designed and modeled fully in SolidWorks, built around a screw-driven sliding jaw. The project covers the sliding jaw mechanism, screw and handle design, base and rail construction, and full assembly documentation including a bill of materials and exploded view.

10 UNIQUE COMPONENTS	16 TOTAL PARTS (INCL. FASTENERS)	1 SCREW-ACTUATED SLIDING JAW	3 FASTENER TYPES
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Design Highlights

- **Screw-driven clamping:** a central jaw screw runs through the base boss and drives the vice jaw along a pair of guide rails, converting rotary handle motion into clamping force.
- **Handle assembly:** a screw bar fitted with ball-shaped bar gobs at each end gives the operator leverage to turn the jaw screw by hand.
- **Sliding rail construction:** twin base plates bolt to the main base to form the guide rails the jaw slides on, with a clamping plate retaining the assembly from above.
- **Documentation:** complete bill of materials, exploded view, and fastener callouts (set screws and an oval fillister head screw) generated directly from the assembly.

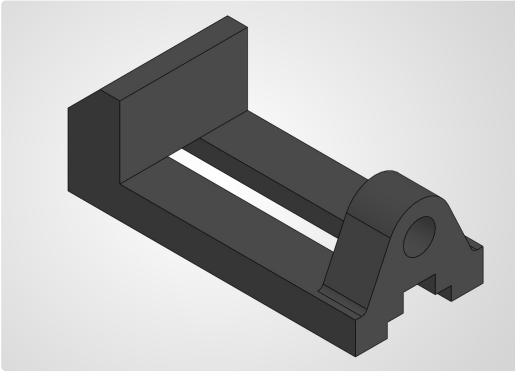
Exploded View



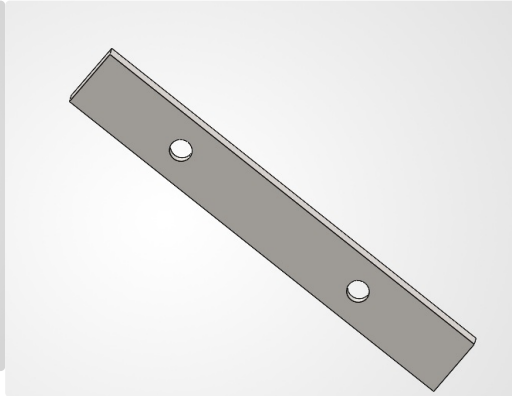
Full exploded assembly — base, base plates, vice jaw, clamping plate, jaw screw, handle, and fasteners

Component Modeling

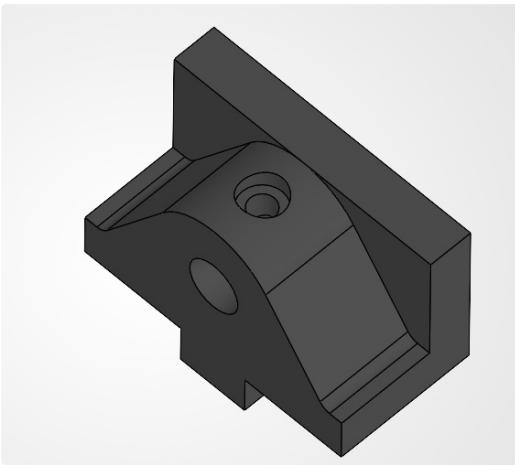
Every part in the assembly was individually modeled as a separate SolidWorks part, from the main base casting to the handle knobs, before being mated together into the working assembly.



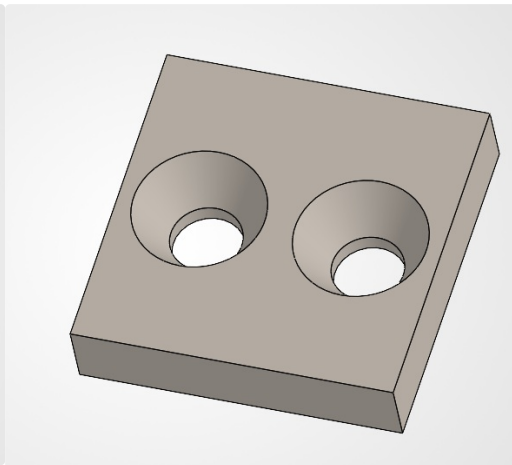
Base — guide channel and boss for the jaw screw



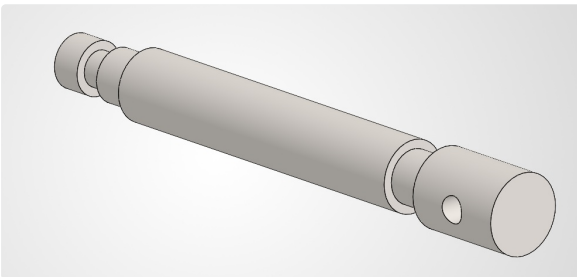
Base plate — bolts to the base to form the sliding rail



Vice jaw — sliding clamp face



Clamping plate — retains the rail assembly



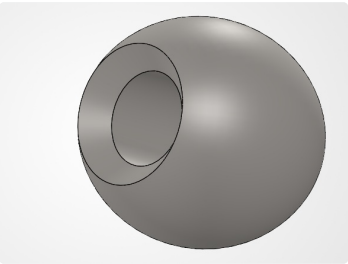
Jaw screw — main actuating spindle with handle cross-hole

Bill of Materials

Extracted directly from the SolidWorks assembly BOM. Every component below was individually modeled as a separate part and mated into the final assembly.

ITEM	PART NUMBER / DESCRIPTION	QTY
1	Base for bench vise	1
2	Base plate for bench vise	2
3	Vice jaw	1
4	Clamping plate	1
5	Set screw 1	4
6	Set screw 2	2
7	Jaw screw	1
8	Screw bar	1
9	Bar gobes	2
10	Oval fillister screw	1

Handle Assembly Detail



Bar gobe — handle knob

Project Overview

A manual drilling jig fixture designed and modeled fully in SolidWorks, built around a toggle-clamp mechanism that locates and secures a cylindrical workpiece for accurate, repeatable hole drilling. The project covers clamp and handle design, guide-bushing placement, and full assembly documentation including a bill of materials and exploded view.

11

UNIQUE COMPONENTS

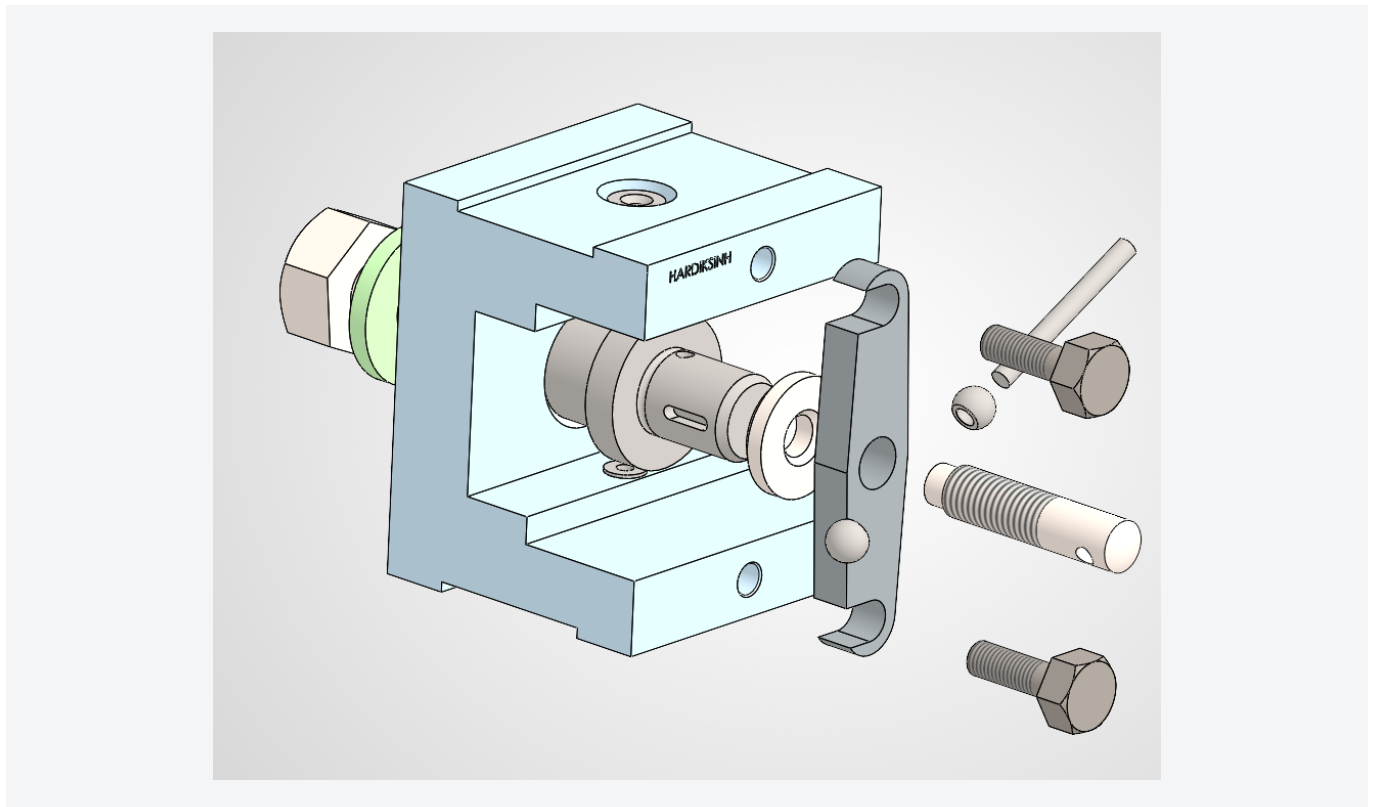
15TOTAL PARTS (INCL.
FASTENERS)**1**TOGGLE-CLAMP
MECHANISM**2**

GUIDE BUSHINGS

Design Highlights

- **Toggle-clamp actuation:** a threaded clamping screw and handle drive the clamp plate against the drill shaft, converting hand-turned rotary motion into a secure hold on the workpiece.
- **Bushing-guided drilling:** two bushings (guide bushings) pressed into the jig body align the drill shaft to hold consistent hole positioning across parts.
- **Serviceable clamp assembly:** the clamp plate is retained with cap screws, a nut, and washer, so the clamping button and drill shaft can be accessed and swapped without reworking the jig body.
- **Documentation:** complete bill of materials, exploded view, and assembled view generated directly from the assembly.

Exploded View



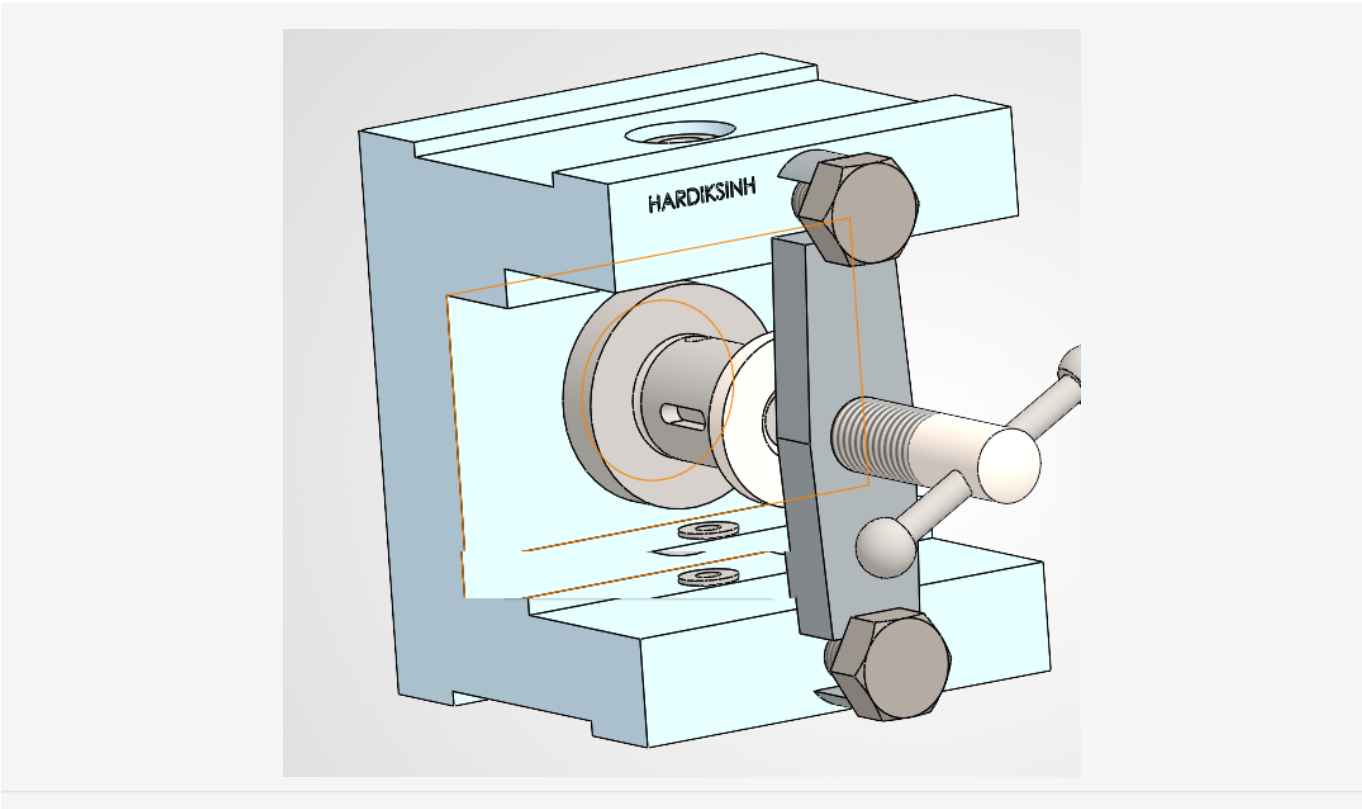
Full exploded assembly — jig body, drill shaft, clamp plate, clamping screw, handle, and fasteners

Bill of Materials

Extracted directly from the SolidWorks assembly BOM. Every component below was individually modeled as a separate part and mated into the final assembly.

ITEM	PART NUMBER / DESCRIPTION	QTY
1	Jig part 1 (body)	1
2	2. Drill shaft	1
3	8. Washer	1
4	11. Nut	1
5	2. Clamp plate	1
6	6. Cap screw	2
7	9. Clamping button	1
8	7. Clamping screw	1
9	4. Handle	1
10	5. Balls	2
11	12. Brushing	2

Assembled View



Mechanical Design Portfolio — Hardiksinh Vaghela
Final assembled drill jig fixture — clamp engaged on the drill shaft

Project Overview

A belt conveyor system designed and modeled fully in SolidWorks, built around a welded frame, dual roller shafts, and a continuous belt for moving parcels along the line. The project covers frame construction, shaft and bearing selection, belt routing, and full assembly documentation including a bill of materials and exploded view.

8

UNIQUE COMPONENTS

37TOTAL PARTS (INCL.
FASTENERS)**2**

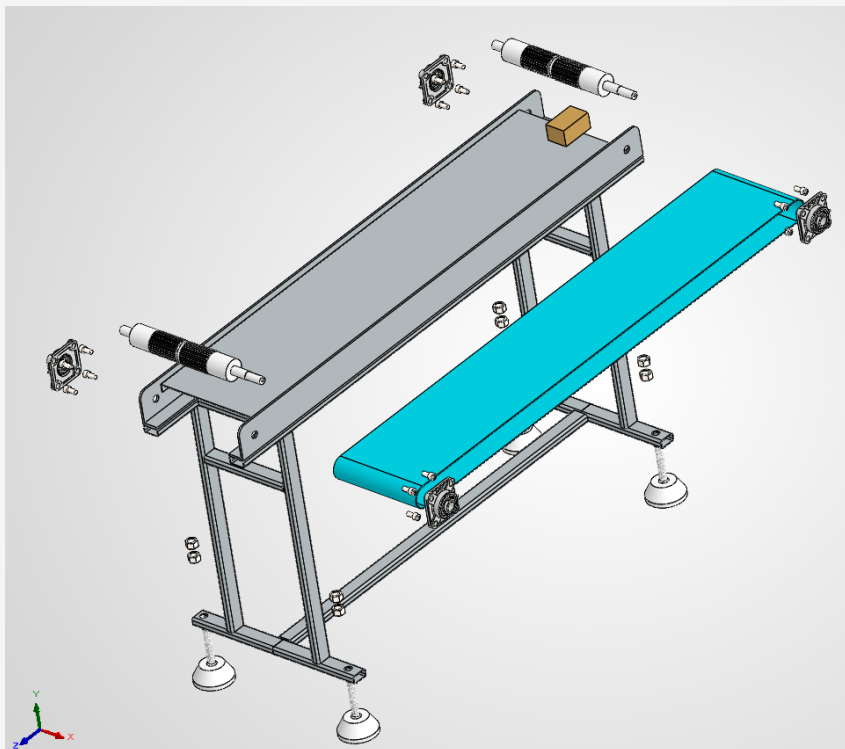
BELT ROLLER SHAFTS

4ADJUSTABLE LEVELING
FEET

Design Highlights

- **Roller-driven belt:** two belt conveyor shafts carry the rollers that drive and return a continuous belt (Belt1-2), moving parcels along the full length of the frame.
- **Bearing-mounted shafts:** SKF UCF 204-12 flange bearing units support both roller shafts at the frame ends for smooth, low-friction rotation.
- **Adjustable leveling feet:** four height-adjustment screws at the base of the frame legs allow the conveyor to be leveled independently on uneven floors.
- **Documentation:** complete bill of materials and exploded view generated directly from the assembly.

Exploded View



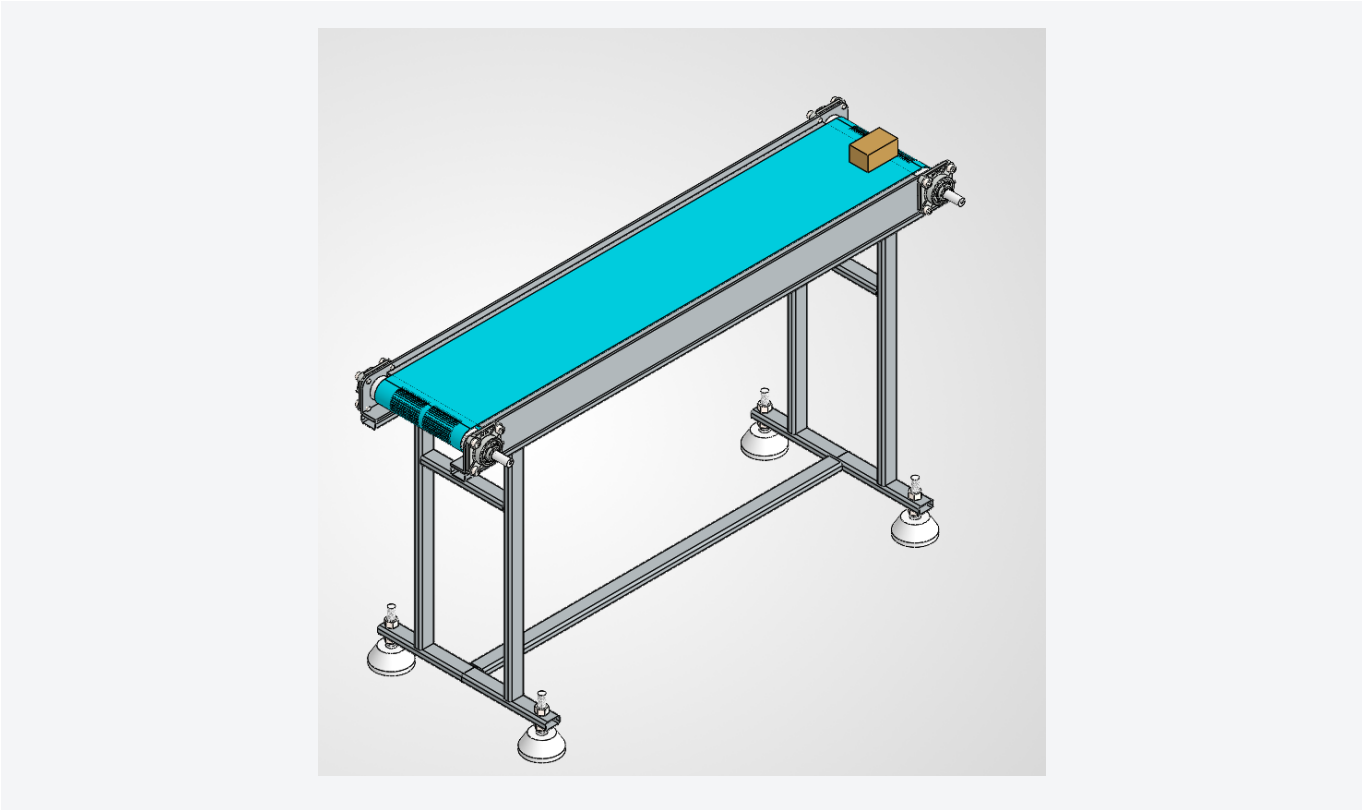
Full exploded assembly — frame, roller shafts, bearing units, belt, and fasteners

Bill of Materials

Extracted directly from the SolidWorks assembly BOM. Every component below was individually modeled as a separate part and mated into the final assembly.

ITEM	PART NUMBER / DESCRIPTION	QTY
1	Conveyor part 1 (frame)	1
2	Belt conveyor shaft	2
3	SKF_UCF 204-12	4
4	Height adjustment screw	4
5	Hexagon Nut ISO 8673 - M16 x 1.5 - W - N	8
6	EN ISO 4762 M10 x 20 - 20N	16
7	Belt1-2	1
8	Parcel	1

Assembled View



Final assembled belt conveyor — parcel shown in transit on the belt